

DoughPlay: An Interactive Projection Game to Facilitate Intergenerational Communication between Grandparents and Grandchildren

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Abstract

Effective grandparent-grandchild communication boosts teens' social skills and older adults' sense of accomplishment while also reducing their isolation. Despite new digital technologies enabling shared activities, a clear "digital divide" remains due to differences in tech familiarity. In this paper, we present DoughPlay, an interactive projection game using multimodal interaction, game-based learning, and traditional Chinese food culture to facilitate intergenerational communication. Our pilot study shows that DoughPlay bridges the intergenerational gap by making intergenerational interactions more equal and enjoyable, easing older adults' tech anxiety, and helping young people to appreciate their grandparents' knowledge and cultural heritage.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**.

Keywords

Intergenerational interaction; Game-based collaboration; Cultural transmission; Interactive projection

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1 Introduction

Intergenerational communication is essential for family cohesion, cultural transmission, and mutual understanding between the generations [1, 7]. However, following accelerating lifestyles, widening generational habits, and increased geographic separation in society, opportunities for intergenerational interaction have declined. Despite the proliferation of digital tools, such as social media, remote video, and virtual reality (VR), a pronounced digital divide remains due to generational differences in technical exposure and proficiency. For example, younger adults can navigate smart devices and applications with ease, whereas older adults often struggle with complex interfaces of these devices [2, 3].

Recent research indicates that VR—by creating immersive virtual environments and offering embodied, interactive narratives [4]—serves as a powerful medium for intergenerational communication. For instance, Shen et al. [7] developed LegacySphere, which combines embodiment, role-play, and storytelling to facilitate VR perspectives and significantly enhance empathic, reflective intergenerational dialogue. Similarly, Zou et al. [9] demonstrated that intergenerational collaboration in a VR environment can effectively reduce ageism. However, VR's potential is tempered by device complexity and safety concerns. Additionally, generational differences in life experience and technical proficiency can hinder mutual understanding and interaction [5].

In this paper, we introduce DoughPlay, a collaborative projection game combining multimodal natural interaction, game-based learning, and traditional Chinese food culture. DoughPlay enables grandparents and grandchildren to cocreate *Huamo* (i.e., flower-shaped steamed buns symbolizing prosperity and heritage) within a virtual space, thus enhancing intergenerational communication, supporting cultural transmission, and reducing digital barriers.

2 Game Design

To achieve this goal, we adopted Ermi's three dimensions of game immersion [8] (sensory, challenge-based, and imaginative), which

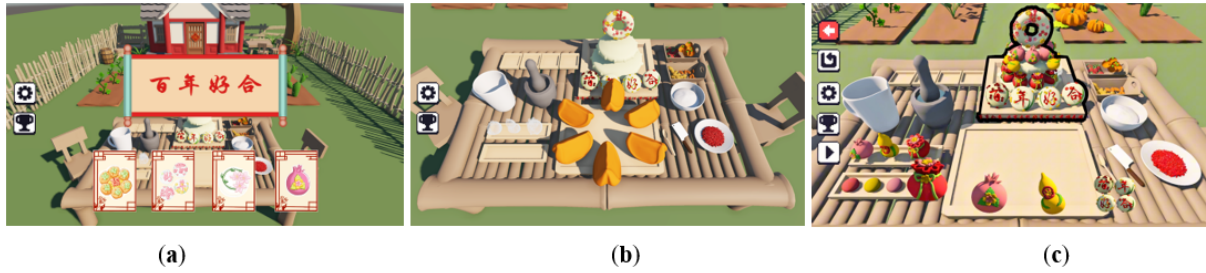


Figure 1: Mini-game interfaces for (a) *Huamo* Matching, (b) *Ingredient Collection*, and (c) *Huamo Making*.

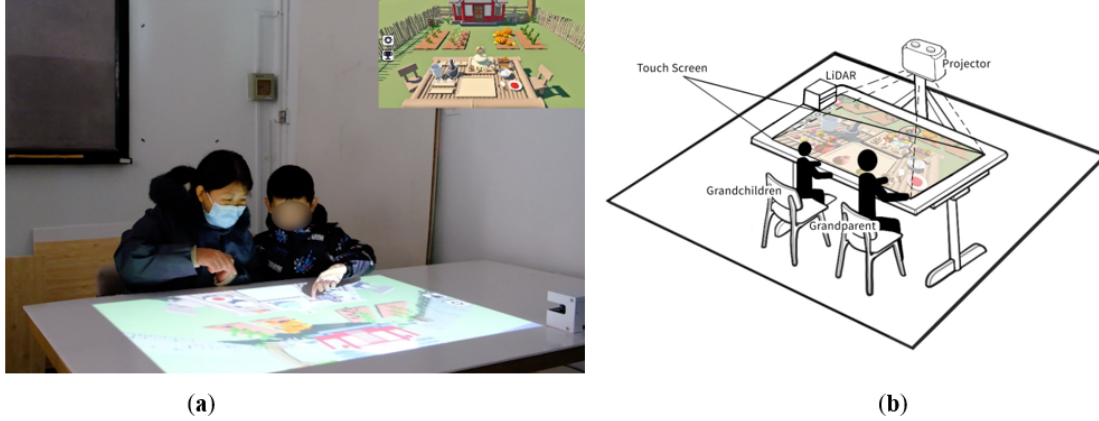


Figure 2: (a) *Gesture interaction in the DoughPlay system* and (b) *overall game layout*.

were applied using three design features in DoughPlay (multi-modal interaction, gamified learning, and a traditional Chinese food-themed context) to support intergenerational engagement and cultural transmission.

2.1 Verbal and Nonverbal Interaction

To ease older users' cognitive load, we simplified game interactions by combining voice prompts with gesture-based inputs. Players can follow voice instructions to complete their tasks and use tapping or dragging gestures to interact with the virtual elements in the projection interface.

2.2 Game-based Progressive Learning

We designed three multilevel mini-games in DoughPlay to support structured collaborations within a traditional *Huamo*-making context: Game 1 (*Huamo Matching*) matches *Huamo* styles to idioms; Game 2 (*Ingredient Collection*) involves collaboration in collecting ingredients; and Game 3 (*Huamo Making*) allows players to replicate or freely create *Huamo* using style cards (Figure 1).

2.3 Culturally Themed Social Context

Traditional Chinese food culture serves as a key medium for family bonding and intergenerational exchange in DoughPlay [3]. We embedded *Huamo*-making as an intangible cultural heritage practice into the mini-games to promote emotional connection and mutual understanding through joint creation[6]. *Huamo* reflects traditional Chinese culinary arts and carries symbolic meanings. In the mini-games, grandchildren learn about the history of *Huamo* and *Huamo*-making skills from older adults, who in turn gain digital

literacy through game interactions, facilitating reciprocal learning and intergenerational exchange.

3 Technical System

The DoughPlay game system was developed using Unity3D within an interactive environment composed of a projector and a LiDAR sensor (Figure 2b). The projector displays the virtual scene onto a physical tabletop surface and when a user touches the tabletop (Figure 2a), the LiDAR sensor captures the touch coordinates and transmits them to enable real-time interaction recognition.

4 User Study and Evaluation

We recruited five grandparents (GP1-GP5, mean $[M] = 63$, standard deviation $[SD] = 2.79$; one man, four women) and their five grandchildren (GC1-GC5, $M = 11$, $SD = 0.84$; two boys, three girls) to participate in a pilot study. The Ethics Review Board of our university approved this study and all participants provided their written informed consent.

Focusing on DoughPlay's usability and user experience, we collected data via (1) video recordings and (2) post-experiment questionnaires and interviews. All participants completed the mini-games and four pairs described the experience as "very interesting." Due to their hearing impairment, GP2 often repeated voice prompts, affecting their experience, while GC2 sought cultural help from GP2 and offered interface support in return. GP5 recalled their childhood memories of making *Huamo* with their family, which sparked more conversations. GP3 found the technology "not difficult" to use, while GC3 described the virtual tabletop as "magical." GP4 felt that DoughPlay brought back memories of her hometown and hoped

her grandchild would inherit this cultural tradition. The participants' suggestions included louder and repeated prompts for older users and more challenges for younger users. Overall, DoughPlay promoted intergenerational collaboration and emotional bonding, while enhancing older adults' confidence in using digital tools through familiar gameplay contexts.

5 Conclusion and Future Work

Our preliminary results suggest that DoughPlay fosters intergenerational collaboration, as demonstrated by the grandparent-grandchild interactions in this study. In the future, we will conduct long-term studies with more participants to refine the DoughPlay system. We also plan to add an ingredient-planting phase to Game 2 to enrich the gameplay flow state and boost engagement.

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